

REQUIREMENTS SPECIFICATION DOCUMENT

E-Commerce Order Management Database System

Week1| Database Management System

Dhivya Grocery Shop

1.Introduction / Overview

Grocery Shop is an online website where customers can buy groceries easily. Customers can view products, add items to the cart, place orders, and make payments. The admin manages products, orders, and stock.

2.Scope

The Grocery Shop website allows customers to register, browse grocery products, place orders, and make payments. The admin can manage products, customers, orders, and inventory.

3.Problem Statement

Nowadays, many people prefer buying skincare and grocery products online because it is convenient and offers more choices. However, finding genuine products from trusted sellers can be difficult, and managing products and customer orders manually is time-consuming for small businesses. Grocery Shop is developed to solve these problems by providing a single online platform where customers can shop with confidence while helping administrators manage products, inventory, and orders efficiently through a well-structured database.

4. Stakeholders

Stakeholder	Responsibilities
Customer	Registers an account, browses products, adds items to the cart, places orders, makes payments, tracks orders, and writes product reviews.
Administrator	Manages products, categories, customer accounts, inventory, discounts, and order details.
Warehouse Staff	Updates stock availability and prepares products for shipping.
Delivery Staff	Delivers orders and updates the delivery status.

5. Functional Requirements

ID	Requirement
FR-01	Customers can create an account and log in securely.
FR-02	Customers can browse skincare and grocery products by category.
FR-03	Customers can search and filter products based on brand, price, or category.
FR-04	Customers can add or remove products from the shopping cart.
FR-05	Customers can place orders and complete payments.
FR-06	Customers can track their orders and view order history.
FR-07	Customers can submit ratings and reviews after purchasing products.
FR-08	The administrator can add, edit, or remove products.
FR-09	The administrator can manage customer information and orders.
FR-10	The system maintains product stock and inventory details.

6. Non-Functional Requirements

Performance

- The website should load quickly.
- Product searches should return results within a few seconds.
- The database should support multiple users without affecting performance.

Usability

- The interface should be simple and easy to understand.
- Customers should be able to complete purchases without difficulty.
- The website should be easy to navigate on both desktop and mobile devices.

Security

- Customer passwords should be stored securely.
- Payment details should be protected.
- Only administrators can modify product and order information.

7. System Modules

Module	Description
Customer Management	Manages customer registration, login, profile, and order history.
Product Management	Adds, updates, deletes, and displays product details.
Category Management	Creates and organizes product categories.
Shopping Cart Management	Manages cart items and calculates the total amount.
Order Management	Handles order placement, tracking, and order history.
Payment Management	Processes payments and generates payment receipts.
Inventory Management	Updates stock levels and monitors product availability.
Review & Rating Management	Allows customers to submit product ratings and reviews.
Admin Management	Manages customers, products, orders, and generates reports.

8. Input Requirements

The system accepts the following inputs:

- Customer registration details
- Login credentials
- Product details
- Category information
- Shopping cart details
- Order information
- Payment details
- Customer reviews and ratings

9. Output Requirements

The system generates the following outputs:

- Customer account confirmation
- Product list based on category or search
- Shopping cart summary
- Order confirmation
- Payment receipt
- Order tracking details
- Customer purchase history
- Product reviews

10. Constraints

- This project is developed for educational purposes.
- Real payment gateway integration is not included.
- Live GPS-based delivery tracking is not supported.
- An internet connection is required to use the website.
- The system focuses on database management and basic e-commerce operations.